# Geometry & Coordinate Geometry

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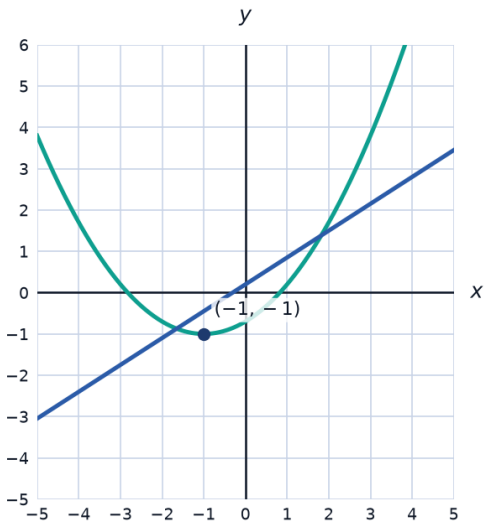
## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** [Foundational] • Target: Distance and midpoint

Find the distance between $(1, 2)$ and $(4, 6)$.

**2** [Foundational] • Target: Distance and midpoint

Find the midpoint of $(2, -3)$ and $(8, 5)$.




**3** [Foundational] • Target: Gradient (slope)

Find the gradient of the line through $(1, 2)$ and $(5, 10)$.

**4** [Foundational] • Target: Gradient (slope)

State whether the line through $(1, 1)$ and $(4, 1)$ is horizontal, vertical, or neither.

**5** [Foundational] • Target: Equations of lines

Find the equation of the line with gradient $2$ passing through $(3, 5)$, in the form $y = mx + c$.

**6** [Foundational] • Target: Equations of lines

Find the equation of the line through $(0, 4)$ and $(2, 10)$.

**7** [Foundational] • Target: Equations of lines

Find the $x$- and $y$-intercepts of the line $2x + 3y = 12$.

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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**8** [Intermediate] • Target: Equations of lines

Write the equation of the line parallel to $y = 4x - 1$ passing through $(2, 9)$.

**9** [Intermediate] • Target: Parallel and perpendicular lines

State the gradient of a line perpendicular to a line with gradient $5$.

**10** [Intermediate] • Target: Parallel and perpendicular lines

Determine whether the lines $y = 2x + 3$ and $y = 2x - 7$ are parallel, perpendicular, or neither.

**11** [Intermediate] • Target: Parallel and perpendicular lines

Find the equation of the line perpendicular to $y = -\frac{1}{3}x + 2$, passing through $(3, 4)$.

**12** [Intermediate] • Target: The equation of a circle

Write the equation of a circle centered at the origin with radius $6$.

**13** [Intermediate] • Target: The equation of a circle

Write the equation of a circle centered at $(2, -3)$ with radius $5$.

**14** [Intermediate] • Target: The equation of a circle

Find the radius and center of the circle $(x - 1)^2 + (y + 4)^2 = 49$.

## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

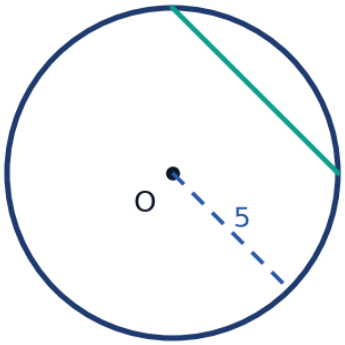
**15** [Challenging] • Target: The equation of a circle • Cross-Lesson: 9.01 Statistics & Probability

Find the equation of the circle centered at the origin that passes through $(3, 4)$.

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**16** [Challenging] • Target: Circles: chords and arcs • Cross-Lesson: 1.01 Algebra & Equations

Points $A$ and $B$ lie on a circle of radius $5$ centered at $O$, with central angle $\angle AOB = 90^\circ$. Find the length of chord $AB$, using $AB = 2r\sin\left(\frac{\theta}{2}\right)$.




**17** [Challenging] • Target: Circles: chords and arcs • Cross-Lesson: 2.01 Functions & Transformations

A chord of a circle with radius $10$ cm is $12$ cm long. Find the distance from the center to the chord.

**18** [Challenging] • Target: Circles: chords and arcs • Cross-Lesson: 3.01 Sequences & Series

Find the circumference of a circle with radius $9$ cm, in terms of $\pi$.

**19** [Challenging] • Target: Polygon angle sums • Cross-Lesson: 4.01 Trigonometry

Find the sum of the interior angles of a hexagon (6 sides).

**20** [Challenging] • Target: Polygon angle sums • Cross-Lesson: 6.01 Limits & Derivatives

Find the measure of each interior angle of a regular pentagon.

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: MCQ — Distance and midpoint

Find the distance between the two points shown.

**a.** $\sqrt{5}$

**b.** $4$

**c.** $3$

**d.** $5$

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**22** [Foundational] • Target: MCQ — Distance and midpoint

Find the midpoint of the segment joining the two points shown.

**a.** $(4, 3)$

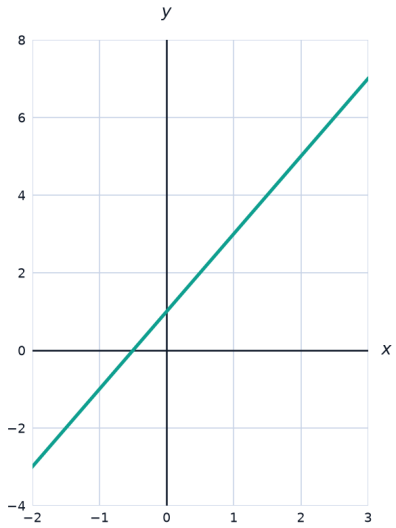
**b.** $(2, 6)$

**c.** $(2, 3)$

**d.** $(8, 6)$

**23** [Foundational] • Target: MCQ — Equations of lines

The graph shows a line. Find its equation in the form $y = mx + c$ by reading the graph.




**a.** $y = \frac{1}{2}x + 1$

**b.** $y = 2x + 1$

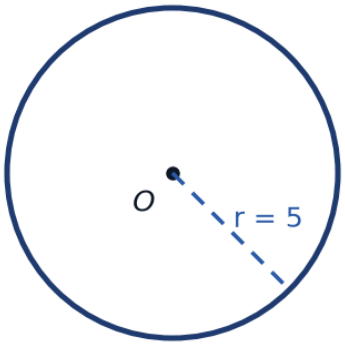
**c.** $y = x + 1$

**d.** $y = 2x - 1$

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**24** [Foundational] • Target: MCQ — The equation of a circle

Write the equation of the circle shown, centered at the origin.




**a.** $x^2 + y^2 = 10$

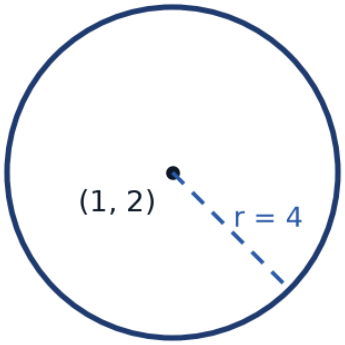
**b.** $x^2 + y^2 = 5$

**c.** $x^2 + y^2 = 20$

**d.** $x^2 + y^2 = 25$

**25** [Foundational] • Target: MCQ — The equation of a circle

The circle shown is centered at $(1, 2)$ with the radius marked. Write its equation.




**a.** $(x - 1)^2 + (y - 2)^2 = 8$

**b.** $(x + 1)^2 + (y + 2)^2 = 16$

**c.** $(x - 1)^2 + (y - 2)^2 = 16$

**d.** $(x - 1)^2 + (y - 2)^2 = 4$

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**26** [Foundational] • Target: MCQ — Circles: chords and arcs

Points $A$ and $B$ lie on a circle of radius $8$ centered at $O$, with central angle $\angle AOB = 60^\circ$. Find the arc length $AB$, in terms of $\pi$.

**a.** $\frac{16\pi}{3}$

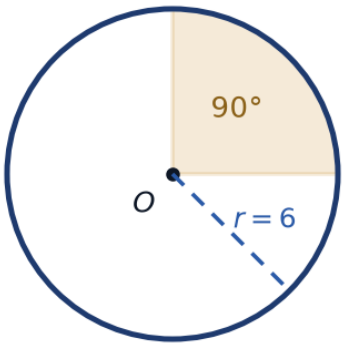
**b.** $\frac{4\pi}{3}$

**c.** $16\pi$

**d.** $\frac{8\pi}{3}$

**27** [Foundational] • Target: MCQ — Circles: chords and arcs

Find the arc length of the $90^\circ$ sector shown, in terms of $\pi$.




**a.** $12\pi$

**b.** $3\pi$

**c.** $1.5\pi$

**d.** $6\pi$

**28** [Foundational] • Target: MCQ — Polygon angle sums

Find the sum of the interior angles of an octagon (8 sides).

**a.** $720^\circ$

**b.** $1{,}440^\circ$

**c.** $900^\circ$

**d.** $1{,}080^\circ$

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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**29** [Intermediate] • Target: MCQ — Polygon angle sums

Find the measure of each interior angle of a regular hexagon.

**a.** $140^\circ$

**b.** $135^\circ$

**c.** $108^\circ$

**d.** $120^\circ$

**30** [Intermediate] • Target: MCQ — Polygon angle sums

Find the sum of the interior angles of a decagon (10 sides).

**a.** $1{,}800^\circ$

**b.** $1{,}260^\circ$

**c.** $1{,}080^\circ$

**d.** $1{,}440^\circ$

**31** [Intermediate] • Target: MCQ — Perimeter and area

A rectangle has length $15$ and width $9$, as shown. Find its perimeter and area.

**a.** $P = 135, A = 48$

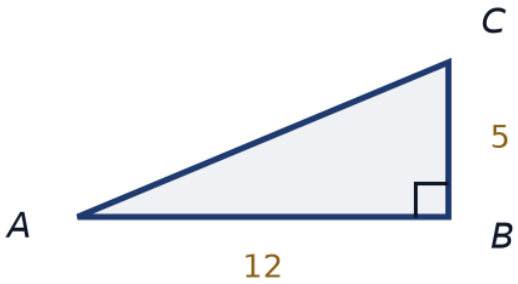
**b.** $P = 48, A = 135$

**c.** $P = 24, A = 135$

**d.** $P = 48, A = 24$

**32** [Intermediate] • Target: MCQ — Perimeter and area

A right triangle has legs $5$ and $12$, as shown. Find its area and hypotenuse.




**a.** $A = 60$, hyp $= 13$

**b.** $A = 30$, hyp $= 17$

**c.** $A = 30$, hyp $= 13$

**d.** $A = 30$, hyp $= 12$

**33** [Intermediate] • Target: MCQ — Perimeter and area

A regular pentagon has side length $8$, with area $A \approx 1.72s^2$. Find its area to the nearest whole number.

**a.** 110

**b.** 124

**c.** 96

**d.** 14

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**34** [Intermediate] • Target: MCQ — Coordinate geometry: identifying shapes

Is the triangle with vertices $A(0, 0)$, $B(6, 0)$, $C(6, 4)$ a right triangle?

**a.** No

**b.** Cannot be determined

**c.** Yes

**d.** Only if it is also isosceles

**35** [Intermediate] • Target: MCQ — Coordinate geometry: identifying shapes

Determine whether the points $A(-3, 2)$, $B(1, 5)$, $C(4, 1)$ form a right triangle, using squared distances.

**a.** Only if it is also equilateral

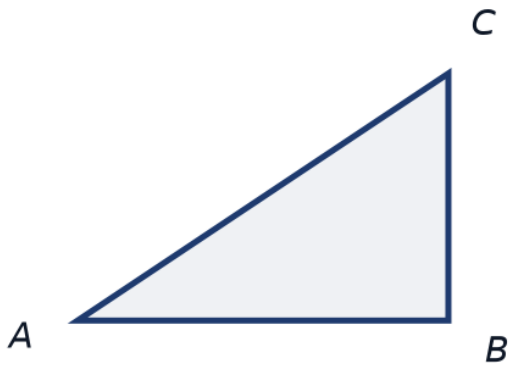
**b.** No

**c.** Cannot be determined

**d.** Yes

**36** [Intermediate] • Target: MCQ — Coordinate geometry: identifying shapes

Is the triangle shown a right triangle?




**a.** Only if it is also isosceles

**b.** Cannot be determined

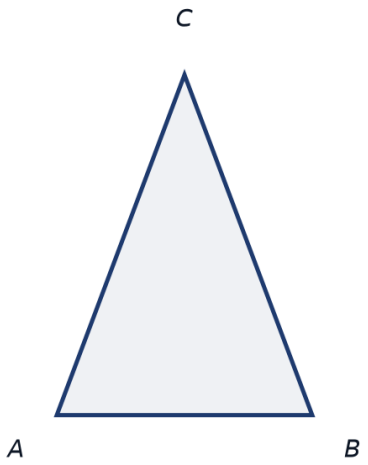
**c.** No

**d.** Yes

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**37** [Intermediate] • Target: MCQ — Coordinate geometry: identifying shapes

Is the triangle shown isosceles, scalene, or equilateral?




**a.** Isosceles

**b.** Cannot be determined

**c.** Equilateral

**d.** Scalene

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: MCQ — Word problems: geometry on a coordinate grid • Cross-Lesson: 3.01 Sequences & Series

A city park is designed on a coordinate grid (in meters). Two gates are located at $(10, 20)$ and $(50, 60)$. Find the distance between the gates, to the nearest meter.

**a.** $113$

**b.** $40$

**c.** $57$

**d.** $80$

**39** [Challenging] • Target: MCQ — Word problems: geometry on a coordinate grid • Cross-Lesson: 4.01 Trigonometry

A circular irrigation sprinkler is centered at $(8, 8)$ on a farm's coordinate grid (in meters) and has a radius of $15$ meters. Write the equation of the circle it waters.

**a.** $(x + 8)^2 + (y + 8)^2 = 225$

**b.** $(x - 8)^2 + (y - 8)^2 = 15$

**c.** $(x - 8)^2 + (y - 8)^2 = 225$

**d.** $(x - 8)^2 + (y - 8)^2 = 30$

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**40** [Challenging] • Target: MCQ — Word problems: geometry on a coordinate grid • Cross-Lesson: 6.01 Limits & Derivatives

A surveyor marks two points on a coordinate map (in km): a base camp at $(3, 4)$ and a summit at $(15, 4)$. Find the midpoint, to be used as a resupply station.

**a.** $(6, 4)$

**b.** $(9, 8)$

**c.** $(9, 4)$

**d.** $(18, 8)$

**41** [Challenging] • Target: MCQ — The perpendicular bisector of a segment • Cross-Lesson: 7.01 Integration & Applications

Find the sum of the interior angles of a nonagon (9 sides).

**a.** $1440^\circ$

**b.** $1620^\circ$

**c.** $1260^\circ$

**d.** $1080^\circ$

**42** [Challenging] • Target: MCQ — Word problems: geometry on a coordinate grid • Cross-Lesson: 8.01 Exponentials & Logarithms

Find the measure of each interior angle of a regular dodecagon.

**a.** $30^\circ$

**b.** $155^\circ$

**c.** $150^\circ$

**d.** $145^\circ$

**43** [Challenging] • Target: Distance and midpoint • Cross-Lesson: 9.01 Statistics & Probability

Find the measure of each interior angle of a regular nonagon.

**a.** $145^\circ$

**b.** $140^\circ$

**c.** $135^\circ$

**d.** $40^\circ$

**44** [Challenging] • Target: Gradient (slope) • Cross-Lesson: 1.01 Algebra & Equations

[illegible] $L$ [illegible] $M$ [illegible] $N$. [illegible] $\alpha$ [illegible] $(3x + 10)^\circ$ [illegible] $\beta$ [illegible] $(5x - 30)^\circ$ [illegible] $x$. In the given figure, transversal line $L$ intersects parallel lines $M$ and $N$. If the measure of angle $\alpha$ is $(3x + 10)^\circ$ and the measure of angle $\beta$ is $(5x - 30)^\circ$, find the value of $x$.

**a.** 20

**b.** 25

**c.** 30

**d.** 35

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**45** [Challenging] • Target: Equations of lines • Cross-Lesson: 2.01 Functions & Transformations

[illegible] $L_1$ [illegible] $L_2$. [illegible] [illegible] [illegible] $L_1$ [illegible] $y = 2x + 3$ [illegible] $L_2$ [illegible] $y = mx + b$[illegible] [illegible] $m$ [illegible] [illegible] [illegible] $b$ [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] $m$[illegible] In the adjacent figure, line $L_1$ is parallel to line $L_2$. If the equation of line $L_1$ is $y = 2x + 3$ and the equation of line $L_2$ is $y = mx + b$, where $m$ is the slope and $b$ is the y-intercept, what is the value of $m$?

**a.** 2

**b.** -2

**c.** 1/2

**d.** -1/2

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